CIUK 2026 · UCL Challenge

# Can we predict co-amoxiclav resistance in E. coli infections?

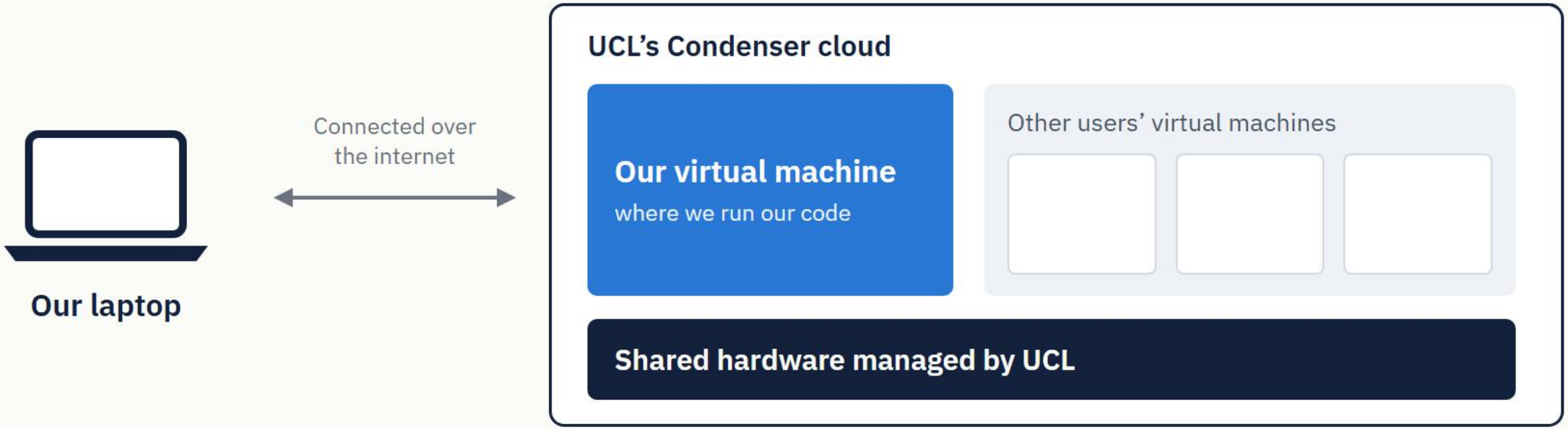
Using decision trees on UCL's Condenser cloud

**UCLuster Too**

Ada Ge · Bobby Ta · Eric Huang · James Loke

**Q1.** What is cloud computing?

# Cloud computing lets us run code on remote computers instead of setting up our own.




Condenser is UCL's private cloud. UCL looks after the physical servers, while we use a virtual machine to run our code. This setup is called Infrastructure as a Service (IaaS).

**Q2.** How did we keep our cloud setup secure?

# We needed to control who could access the machine, protect our connection and keep our keys private.




We kept passwords and private keys out of our public GitHub repository.

We accessed the VM through JupyterHub. Our SSH key was stored there with access restricted to our team.

The VM ran in UCL's facilities, isolated from other users' virtual machines.

Condenser isn't certified for sensitive patient data. If we wanted to work with real medical records, we'd need additional safeguards and an approved environment.

**Q3.** Why did we change the file permissions?

# Anyone using the machine could originally read our data file. We fixed that.

**Before — File readable by other users**

| | Read | Write |
|---|---|---|
| Our account | ✓ | ✓ |
| Our group | ✓ | X |
| **Other users** | ✓ | X |

Changed permissions using chmod

**After — Access restricted to the file owner**

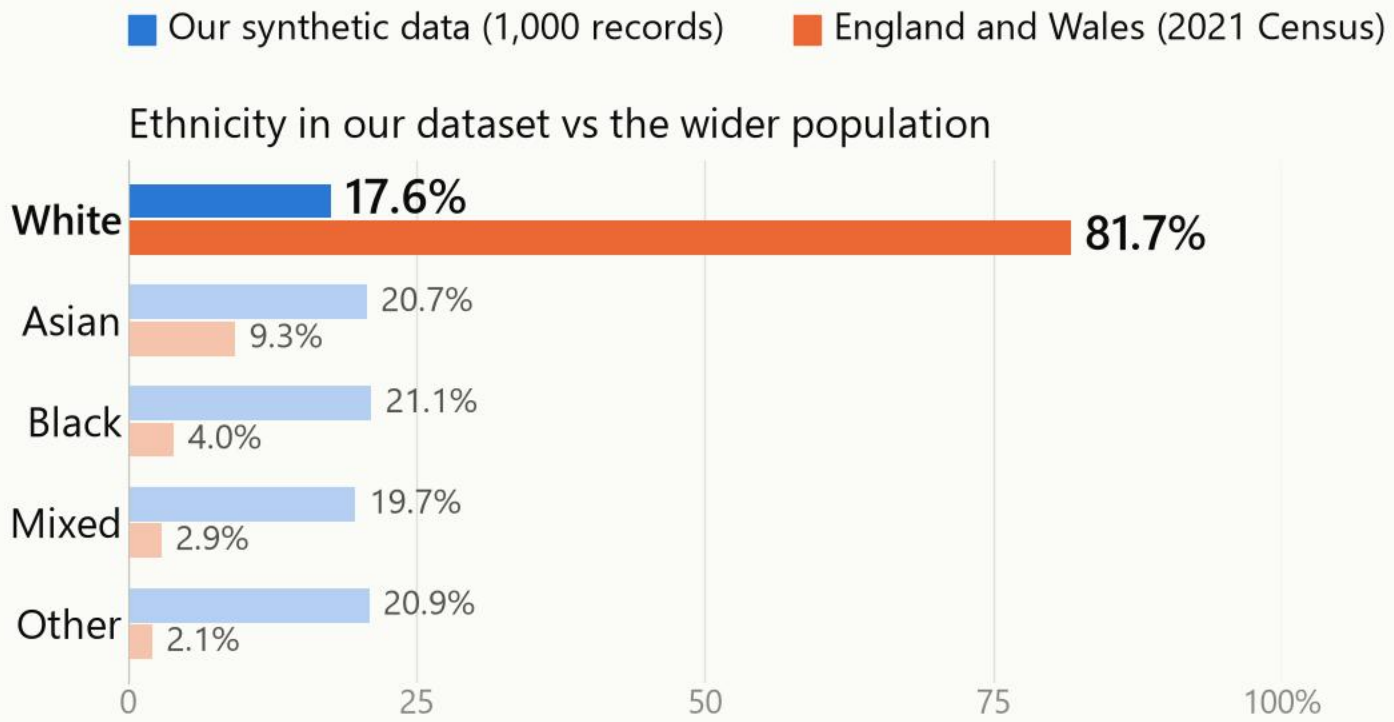
| | Read | Write |
|---|---|---|
| Our account | ✓ | ✓ |
| Our group | X | X |
| **Other users** | X | X |

```
-rw-r--r--. 1 cc cc 94198 Oct 26 16:44 dig_health_hub_amr.csv
$ chmod 700 data
$ chmod 600 data/dig_health_hub_amr.csv
-rw-------. 1 cc cc 94198 Oct 26 16:44 dig_health_hub_amr.csv
# We also set umask 027 to stop new files being world-readable.
```

**Q4.** Why does representative training data matter?

# We can’t assume a model will work for patients who aren’t represented in its training data.

Our 1,000 synthetic records look very different from the population of England and Wales.
They also only cover people aged 20–79.







We have no data for anyone younger than 20 or older than 79, so we can’t tell how well our model would work for them.

**Q5.** Why use synthetic data instead of real patient records?

# Synthetic data lets us test our models without using anyone's real medical records.

## A real patient's record

| | |
|---|---|
| Name | [redacted] |
| NHS number | [redacted] |
| Date of birth | [redacted] |
| Home postcode | [redacted] |
| Diagnosis | |
| Test results | |

**These details belong to a real person. Their medical information is protected by UK GDPR, and we would need a lawful basis and proper safeguards to use it.**

## A synthetic record, like the ones we used

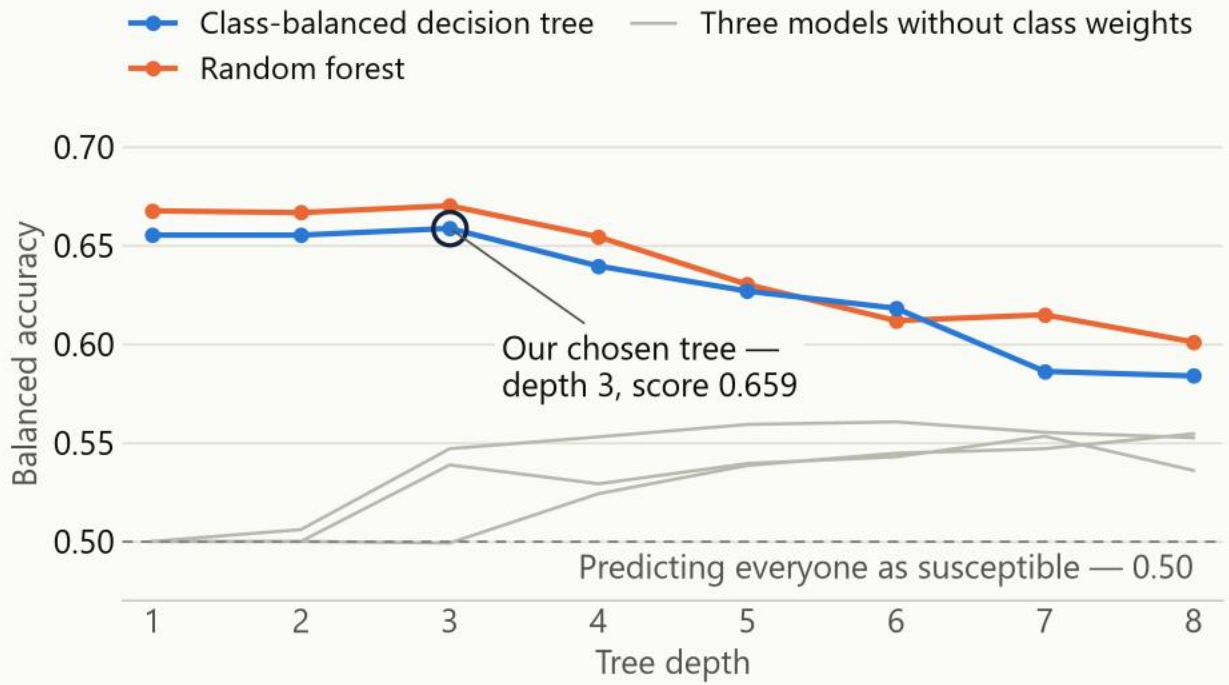
The details are generated, including the patient ID and date of birth.

| | |
|---|---|
| Patient ID | |
| Date of birth | |
| Sex and ethnicity | |
| Deprivation | |
| NHS trust | |
| Diagnosis code | |
| Test results | |

**The record doesn't identify a real patient, so the privacy risks are much lower, even when working on shared computing infrastructure.**

**Q6.** Why did we choose this decision tree?

# Depth 3 worked best for a single tree. Making it deeper didn't help.




### Predicting everyone as susceptible

76% standard accuracy, but only 0.50 balanced accuracy.

### Our depth-3 decision tree

Balanced accuracy: **0.659**

### Random forest — 200 trees

Balanced accuracy: **0.670**

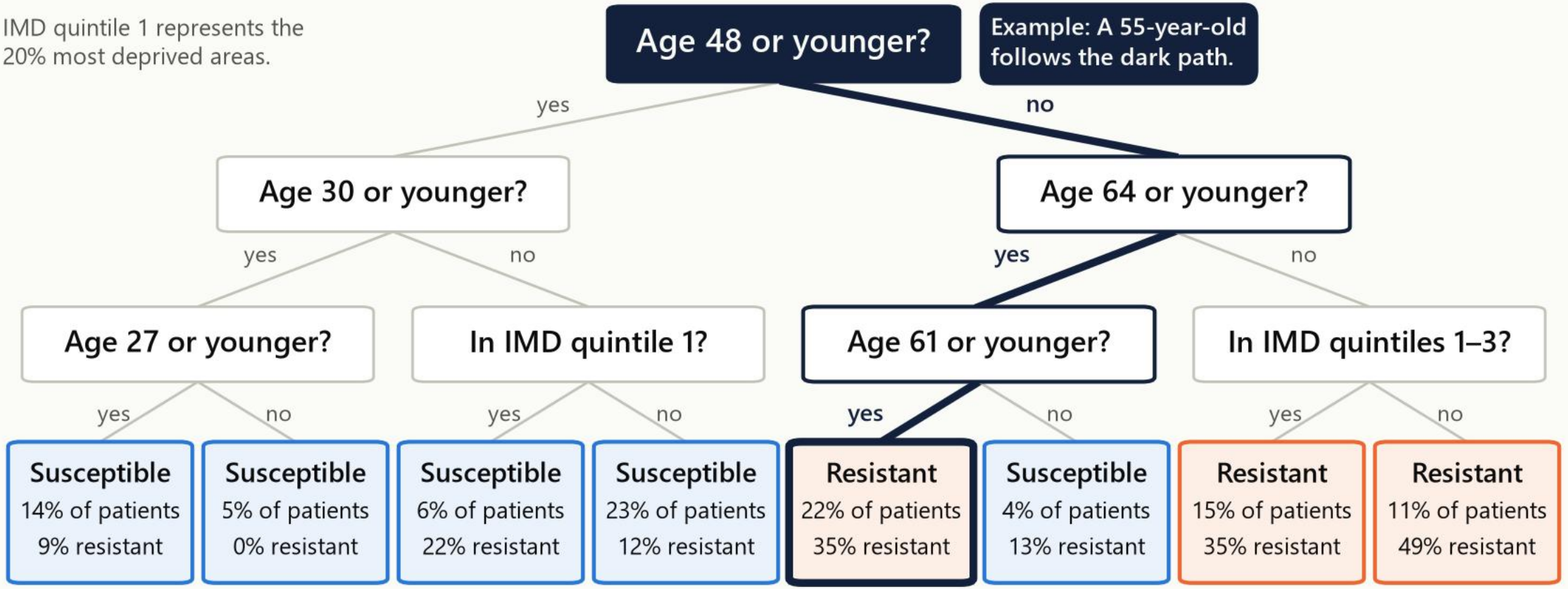
The forest scored slightly higher, but the single tree is much easier to understand and explain.

**Q6.** Why did we choose this decision tree?

# Here's how our final decision tree makes its predictions.

Age does most of the work. Our tree predicts resistance for patients aged 49–61 and 65 or older. The other ages are predicted susceptible.

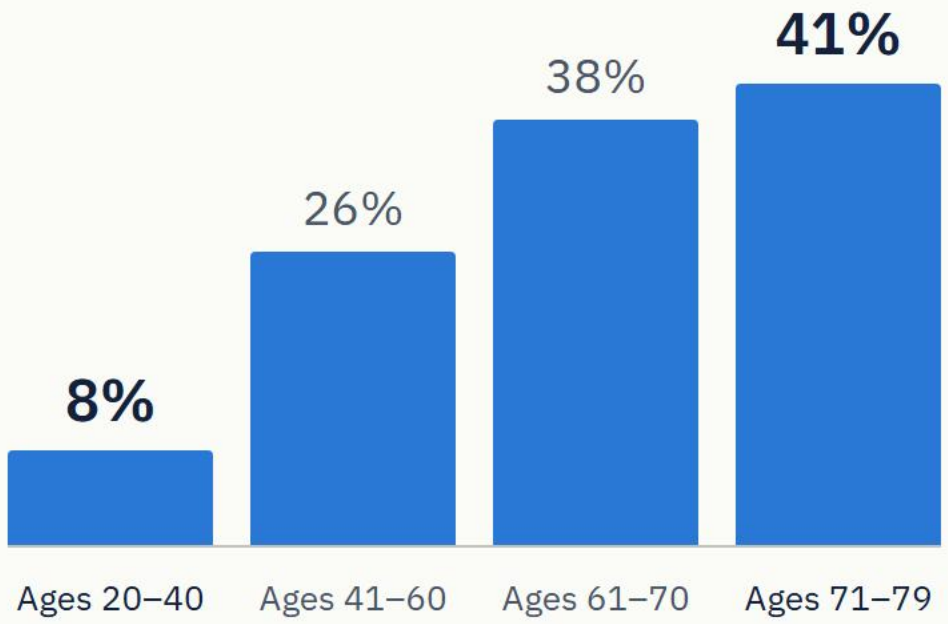
IMD quintile 1 represents the 20% most deprived areas.



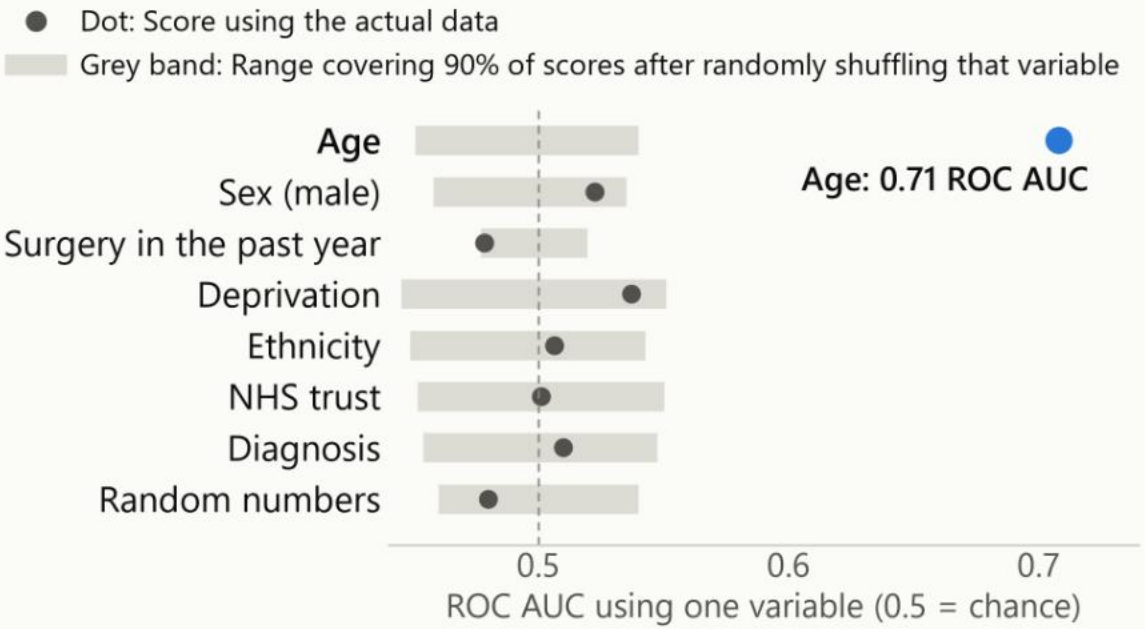
**Q6.** Why did we choose this decision tree?

# Resistance became more common with age.
# We didn't find much useful information in the other variables.

### Percentage resistant in each age group




### How useful was each variable on its own?



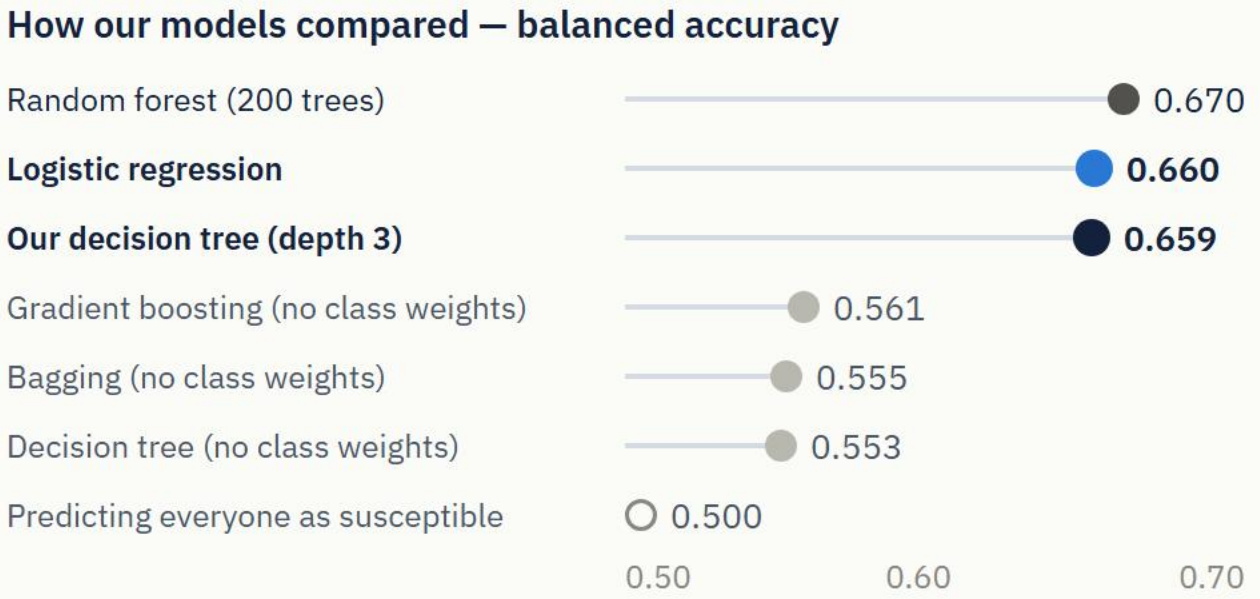
**Q7.** What other models did we try?

# Logistic regression performed almost exactly as well as our tree, and helped us understand the age effect.

## 1.5×

For every additional 10 years of age, the odds of resistance were about 1.5 times higher.

Logistic regression can model a gradual increase in risk, while a decision tree divides patients into groups. With just 1,000 records and age providing most of the useful information, we didn't see a strong reason to use something more complicated, like a neural network.



**Q8.** How did we work together?

# We split the work into two pairs and kept each other updated every day.

### Cloud setup and modelling

**Bobby** — Kept the cloud machine up to date, tested a fresh install from our README and ran the final submission checks.

↕

**James** — Developed the modelling script.

### Analysis and presentation

**Ada** — Set up the first VM environment and README, explored the data and chose how to evaluate the models.

↕

**Eric** — Led the video and slides. Tested every risk factor against random noise and ran the logistic regression.

A quick 15-minute catch-up every evening.

We kept our plan, role guides and work-session notes together in one GitHub repository.

**Q8.** How did we work together?

# We ran into three problems along the way. Here's how we dealt with them.

**Problem 1**

Only one of us received the login details, and the username was missing.

↓

**How we solved it**

We followed the organisers' instructions to share access and found the missing username in their access guide.

**Problem 2**

Git wasn't installed on the cloud machine.

↓

**How we solved it**

We made changes on our laptops, pushed them to GitHub and used `curl` to fetch the files on Condenser.

**Problem 3**

Our first model predicted that every patient was susceptible.

↓

**How we solved it**

We realised ordinary accuracy was hiding the problem, so we switched to balanced accuracy and added class weights.

# Age was the clearest predictor we found in our synthetic dataset.

Our final model was a depth-3 decision tree with class balancing. It achieved 0.66 balanced accuracy while keeping its predictions easy to explain.

These results demonstrate our modelling approach, but aren’t intended for clinical use.

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`github.com/adellama75/cluster-challenge-ucl`